

Q1:

A university uses a Smart Library System to manage book borrowing activities with interactions between students, librarians, library servers, and the notification system. The process flows as follows:

A student scans their ID card to access the system. If the card is invalid or not registered, access is denied. After three invalid attempts, the librarian is notified, and access is blocked. If the card is valid, the system authenticates the student.

The system displays available books and allows the student to search. If a book is selected, the system checks the availability. If unavailable, the student is added to a waiting list, and a notification is sent when the book is returned.

If available, the student requests to borrow the book, and the system verifies borrowing limits. If limits are exceeded, the system denies the request and alerts the student. If within limit, the system processes the request, updates the book inventory, and generates a digital receipt.

The librarian is notified of the transaction, and the due date is scheduled. If the book is not returned by the due date, the system sends a reminder, and if still unreturned after 7 days, a fine is applied and admin is notified.

Once the book is returned, the system updates the record, and if the book was on a waiting list, the next student is notified automatically.

Design a Sequence Diagram for the above scenario.